

Predicting Diabetes Progression

Machine Learning Project · Luca Frittitta



GradientBoostingRegressor

GridSearchCV

scikit-learn

RidgeCV Baseline

Predict the 1-year progression of diabetes using 10 standardized clinical variables, supporting a healthcare organization in prevention and monitoring of chronic diseases.

442

Patients

10

Clinical Features

1 yr

Prediction Horizon

R² GBR: 0.47 /
Ridge: 0.49

Models Scores

APPROACH

- 01 Establish a Ridge Regression baseline to set the minimum performance threshold
- 02 Train a Gradient Boosting Regressor
- 03 Tune hyperparameters via GridSearchCV (5-fold CV)
- 04 Evaluate with R², MAE, RMSE on test set
- 05 Analyze feature importance for clinical insights

FEATURES USED

Feature	Type	Notes
age	Numeric	Age in years (standardized)
sex	Binary	Sex indicator (standardized)
bmi	Numeric	Body Mass Index — top predictor
bp	Numeric	Average blood pressure
s1–s6	Numeric	Blood serum indicators
target	Numeric	Disease progression at 1 year

Correlation Heatmap — Feature Relationships



bmi & s5 → target

Highest correlation (~0.59) — strongest clinical predictors

s1 & s2 correlation

"Mutually correlated (0.90) — multicollinearity affects linear models but not GBR. All features retained."

s3 (HDL) negative

Negative with target (-0.39) — HDL is protective

sex & age low

Low correlation with target — limited predictive power alone

Which clinical variables drive the prediction?



~39%

bmi — Body Mass Index

~29%

s5 — Log-Triglycerides

~13%

bp — Blood Pressure

★ Together: ~81% of predictive power

Evaluation Metrics & Model Performance GBR

~0.47

R² GBR Score

~43.5

GBR MAE

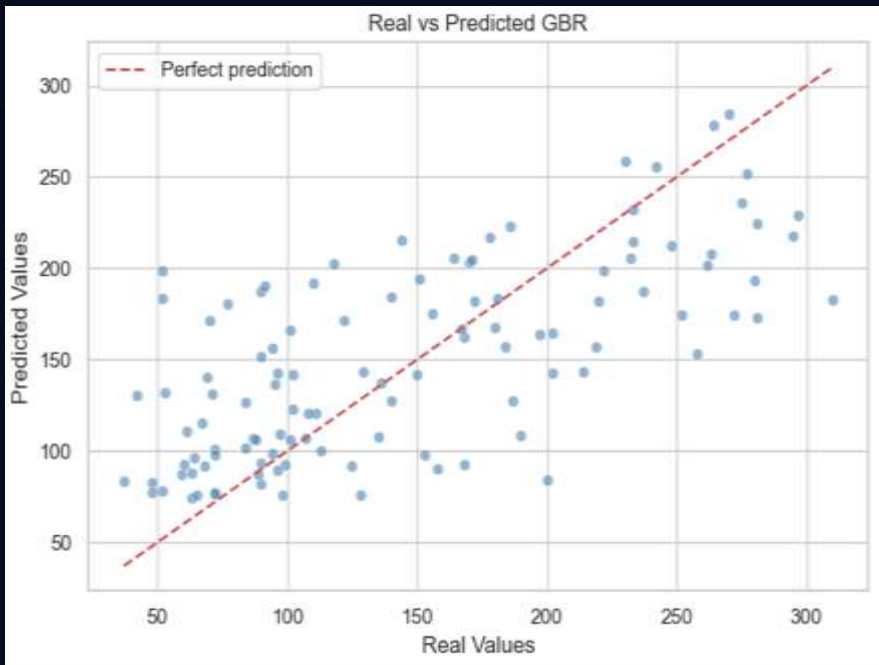
~54.8

GBR RMSE

0.42 ±0.12

GBR CV R² (5-fold)

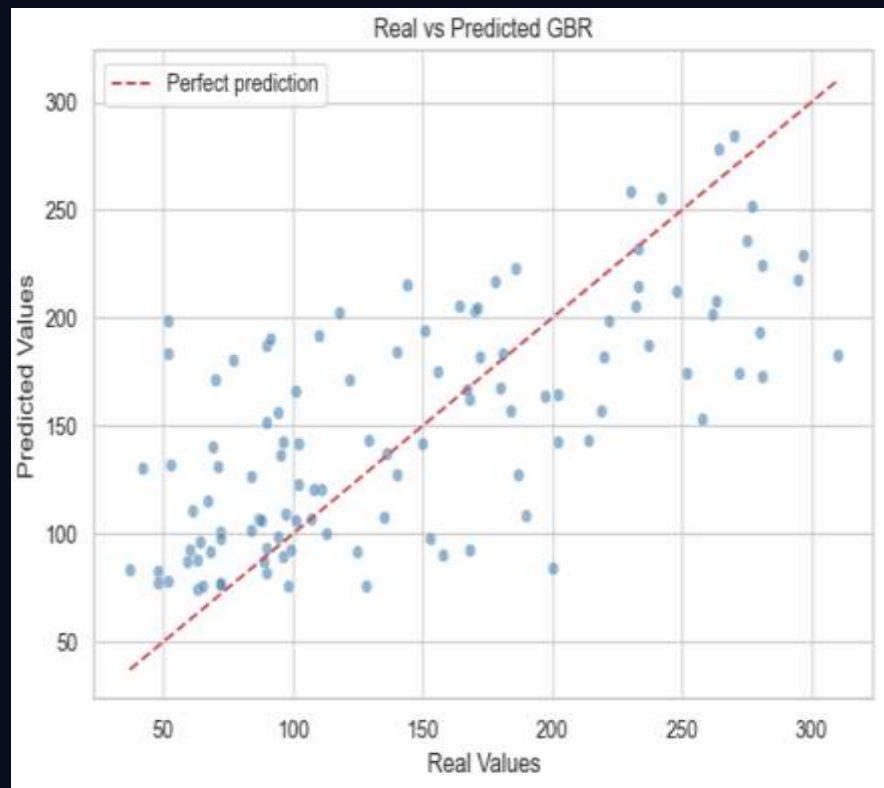
The model is **useful for identifying general risk trends** (low/medium/high), but not precise enough to support individual clinical decisions without further validation. An R² of 0.47 accurately reflects this visual dispersion. Patient variability depends on genetics, lifestyle and therapy not captured in the dataset.



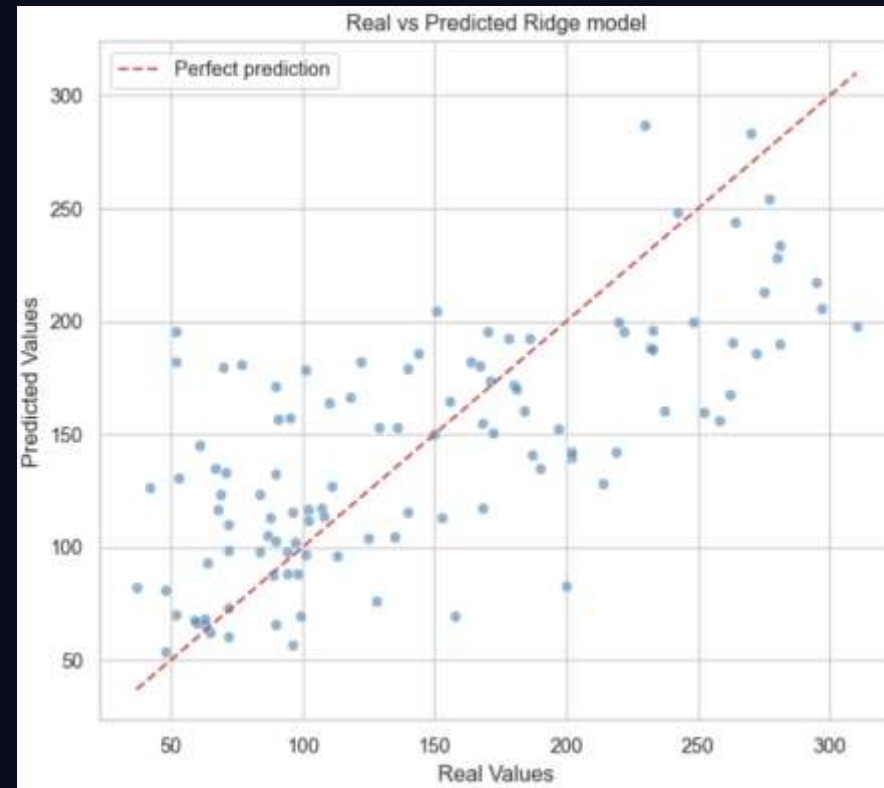
KEY OBSERVATIONS

- ✓ **Clear positive trend** — points follow the direction of the red dashed diagonal, confirming the model has learned the relationship between features and the target variable.
- ✓ **Good performance in the central range (100–200)** — in this range, points cluster closer to the perfect prediction line, where the majority of patients are concentrated.
- ✗ **High variance for severe progressions** — in the 230–310 range, predictions spread widely (e.g., from ~155 to ~285 for a real value of ~260), meaning the model struggles to distinguish the most severe cases

Model Comparison — Ridge Regression vs Gradient Boosting



The model comparison shows that Ridge Regression matches and marginally exceeds GBR on all three metrics. This is not a failure of the ensemble method — it reflects the structural limits of a small dataset. GBR's advantage emerges with larger samples and stronger non-linear patterns, neither of which is present here.



	Ridge (Baseline)	GBR
R^2	0.49	~0.47
MAE	41.55	~43.5
RMSE	53.01	~54.8

Key Findings for the Medical Team

🏆 **On this dataset, Ridge Regression slightly outperforms GBR (R^2 0.49 vs 0.47)**

With only 442 samples and predominantly linear relationships, the simpler model generalizes better. GBR remains valid when non-linear patterns or larger datasets are present.

📊 **BMI is the top predictor (~39%)**

High BMI patients need priority monitoring — it's the single strongest signal of disease progression.

📊 **s5 (log-triglycerides) is #2 (~29%)**

Elevated lipids signal accelerated insulin resistance. High s5 = high-risk patient flag.

❤️ **Blood pressure ranks 3rd (~13%)**

Consistent with cardiovascular complications in diabetes — bp monitoring is essential for all patients.